230701104

EXPERIMENT – 3

Performing Man-in-the-Middle (MITM) Attacks Using Ettercap in Kali Linux

Aim

To perform and analyze a Man-in-the-Middle (MITM) attack using the Ettercap tool in Kali Linux for educational and ethical hacking purposes.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Security Tool:** Ettercap
- **System Requirements:**
  - RAM: Minimum 4 GB (8 GB recommended)
  - Disk Space: Minimum 30 GB
  - Processor: Intel/AMD with Virtualization support

Introduction

A Man-in-the-Middle (MITM) attack is a type of cyberattack where an attacker secretly intercepts and possibly alters communication between two parties without their knowledge. Ettercap is a powerful open-source network security tool used for MITM attacks, traffic interception, and protocol analysis.
Using Kali Linux within Oracle VirtualBox provides a controlled and isolated environment to study MITM attacks ethically without affecting real-world systems.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Launch Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Verify Network Connectivity

1. Ensure Kali Linux is connected to the same network as the target system.
2. Check IP address using:
3. ifconfig

Step 3: Launch Ettercap

1. Open Terminal in Kali Linux.

2. Start Ettercap with graphical interface:
3. ettercap -G

Step 4: Select Network Interface

1. In Ettercap GUI, click **Sniff → Unified Sniffing**.
2. Select the active network interface (e.g., eth0 or wlan0).
3. Click **OK**.

Step 5: Scan for Hosts

1. Click **Hosts → Scan for Hosts**.
2. After scanning, select **Hosts → Hosts List**.
3. Identify the **target system** and **gateway**.

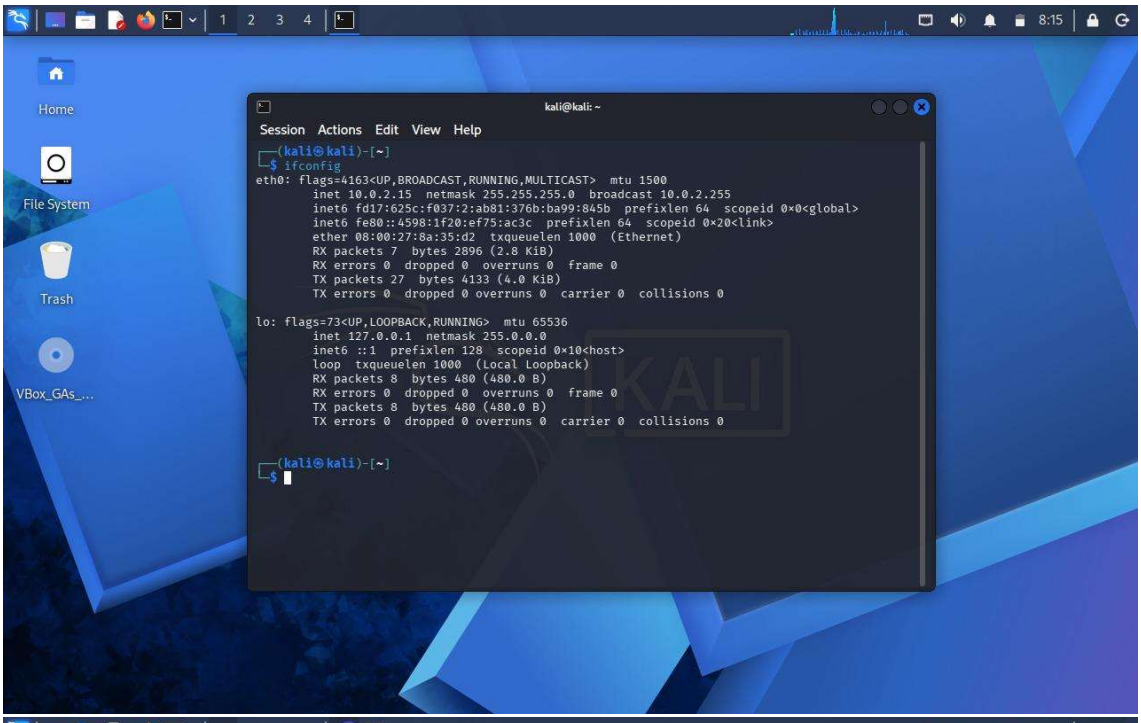
Step 6: Configure MITM Attack

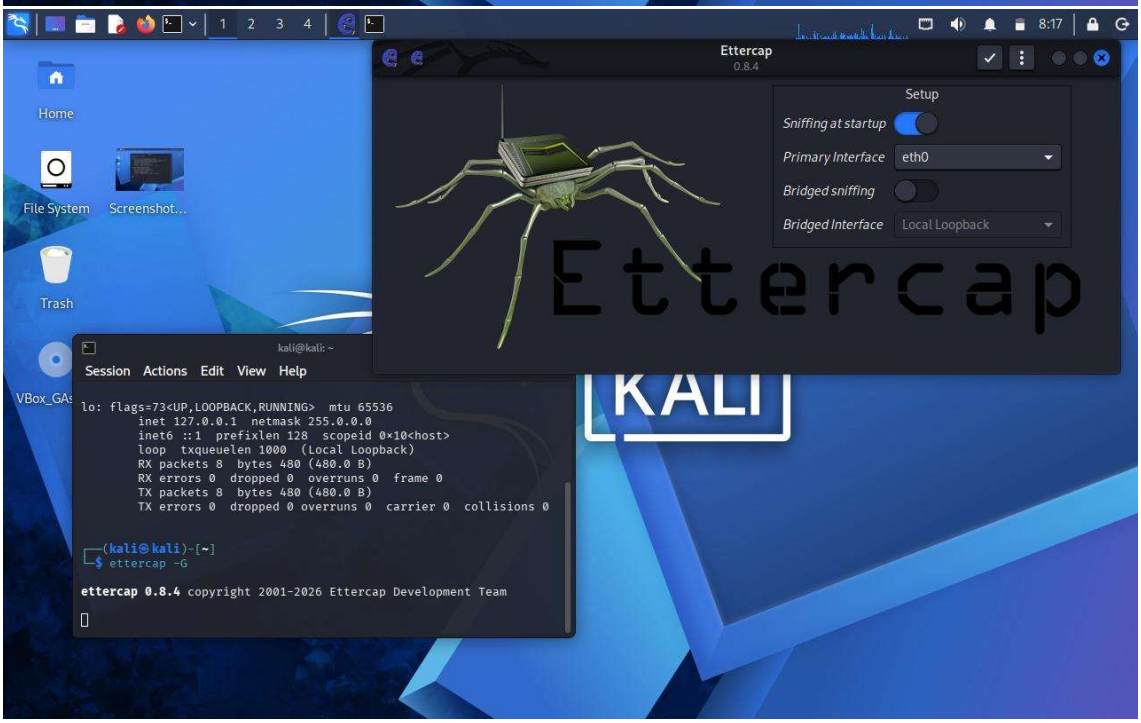
1. Add the target system as **Target 1**.
2. Add the gateway/router as **Target 2**.
3. Click **MITM → ARP Poisoning**.
4. Enable **Sniff remote connections**.
5. Click **OK**.

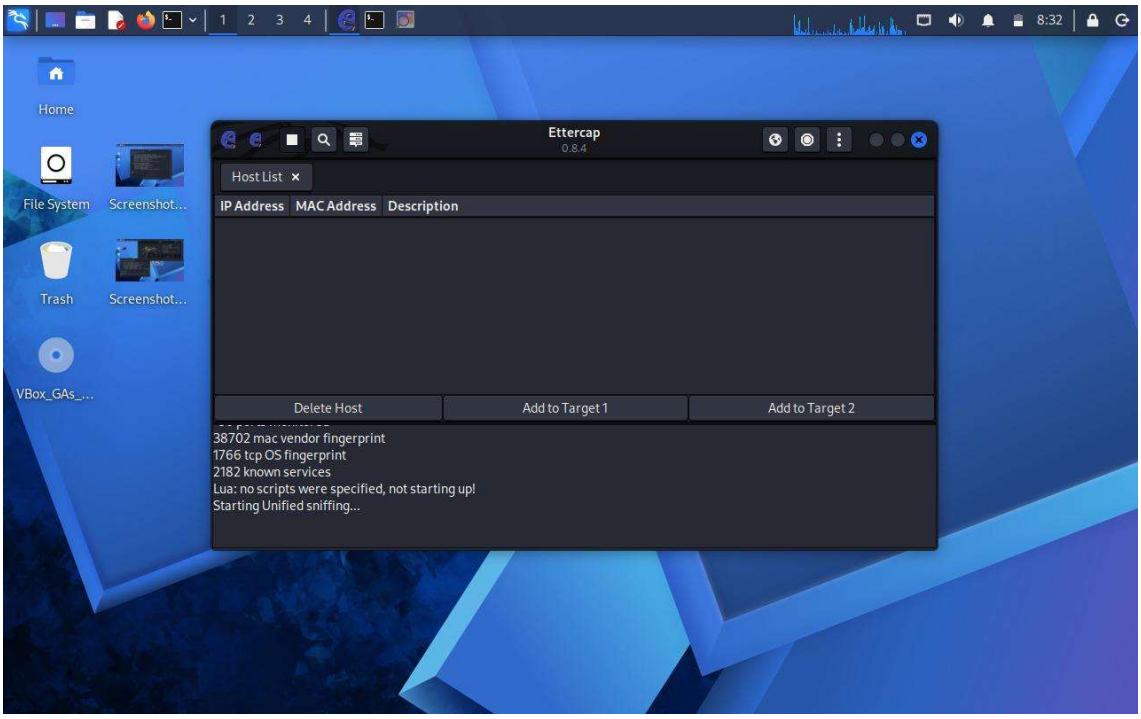
Step 7: Start Sniffing

1. Click **Start → Start Sniffing**.
2. Observe intercepted traffic in real time.

Output


kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
inet6 fd17:625c:f037:2:ab81:376b:ba99:845b prefixlen 64 scopeid 0x0<global>
inet6 fe80::4598:1f20:ef75:ac3c prefixlen 64 scopeid 0x20<link>
ether 08:00:27:8a:35:d2 txqueuelen 1000 (Ethernet)
RX packets 7 bytes 2896 (2.8 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 27 bytes 4133 (4.0 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 8 bytes 480 (480.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 8 bytes 480 (480.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali㉿kali)-[~]
$


Ettercap
0.8.4
Setup
Sniffing at startup
Primary Interface
eth0
Bridged sniffing
Bridged Interface
Local Loopback
Ettercap
kali@kali: ~
Session Actions Edit View Help
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 8 bytes 480 (480.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 8 bytes 480 (480.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali㉿kali)-[~]
$ ettercap -G
ettercap 0.8.4 copyright 2001-2026 Ettercap Development Team



Result

The MITM attack was successfully performed using Ettercap in Kali Linux.
The experiment demonstrated how insecure network communication can be intercepted, highlighting the importance of encryption and secure network protocols.